

PROJECT DESIGN PHASE – II

Solution Requirements (Functional & Non-functional)

Date: 30 September 2026

1. Project Details

Team ID	SWTID-2026-3350
Project Name	PocketSmart AI – Your Smart Budget & Recommendation Assistant
Team size / Members	4 – Esther Ammo M , Epsipa R , Eswari M , Fayisa Fathima A

2. Functional Requirements

Following are the functional requirements of the proposed PocketSmart AI solution. They follow the Functional Requirement (Epic) and Sub Requirement (Story / Sub-Task) structure used in the provided template.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Profile & Financial Input	Enter user information, income, budget amount, expense details, expense categories and financial preferences. Validate required financial input using FastAPI/Pydantic.
FR-2	Budget & Expense Analysis	Process income, budget and expense information. Calculate spending totals and remaining budget and display a clear financial summary.
FR-3	Personalized AI Recommendations	Generate personalized budgeting and spending recommendations using Gemini AI. Display recommendations and explanatory insights with the financial summary.
FR-4	Recommendation / Budget Update	Accept user feedback or updated financial information such as changed budget, expenses or spending preferences. Generate updated recommendations based on the latest information.
FR-5	User & Financial Data Database	Store user details, budgets, expenses and generated recommendations in SQLite using

		SQLAlchemy. Preserve relevant records for later viewing.
FR-6	Admin View	View registered users and stored budgeting information. View relevant budget, expense and recommendation records according to access controls.
FR-7	Web Interface	Provide Jinja2 pages for user input, budget/expense results and recommendation views. Render AI-generated content dynamically in the web pages.
FR-8	Backend Routing & Deployment	Implement FastAPI routes for home, financial input, budget analysis, recommendation generation, feedback/update and user view. Run and verify the application locally using Uvicorn.

3. Non-functional Requirements

Following are the non-functional requirements of the proposed PocketSmart AI solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface should be simple and easy to use for entering financial details, viewing budget summaries and recommendations, and submitting updates or feedback.
NFR-2	Security	User financial information and Gemini API configuration should be handled securely; sensitive configuration and API credentials should not be exposed in the user interface or source code.
NFR-3	Reliability	The application should validate inputs, handle errors gracefully, preserve stored financial and recommendation data, and avoid losing valid records during normal operation.
NFR-4	Performance	Budget analysis and recommendation requests should be processed efficiently and results should be displayed without unnecessary delay.
NFR-5	Availability	The PocketSmart AI web application should be available whenever the configured FastAPI/Uvicorn service is running.

NFR-6	Scalability	The application structure should allow additional users, budgeting features, recommendation capabilities, AI prompts and database records to be added in future.
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4. Requirements Traceability Summary

Feature Area	Functional Requirement	Related NFRs
User Input	FR-1	NFR-1, NFR-4
Budget & Expense Analysis	FR-2	NFR-3, NFR-4
AI Recommendations	FR-3	NFR-1, NFR-4
Feedback & Recommendation Update	FR-4	NFR-3, NFR-4
Database	FR-5	NFR-2, NFR-3, NFR-6
Admin View	FR-6	NFR-1, NFR-2
Web Interface	FR-7	NFR-1, NFR-5
Backend & Deployment	FR-8	NFR-3, NFR-4, NFR-5, NFR-6